

Requirements		
Req ID	Requirement Title	Requirement Intent
ACC-REQ-001	System Initialization	On system start, the controller shall enter the Power_Off with compressor_on = 0, fan_on = 0, fault_indicator = 0, and status outputs reset.
ACC-REQ-002	Temperature Setpoint Input	The controller shall use set_temperature as the desired room temperature reference for cooling control.
ACC-REQ-003	Power ON Command	When power_button == 1, the controller shall enter the enabled operating state, compute temperature_error = room_temperature - set_temperature, and determine cooling demand based on temperature_error and temperature_hysteresis_band.
ACC-REQ-004	Power OFF Command	When power_button == 0, the controller shall enter the Power_Off state and set compressor_on = 0 and fan_on = 0.
ACC-REQ-005	Cooling Demand Activation	If temperature_error >= temperature_hysteresis_band, the controller shall set cooling_demand = 1.
ACC-REQ-006	Cooling Demand Deactivation	If temperature_error <= -temperature_hysteresis_band, the controller shall set cooling_demand = 0.
ACC-REQ-007	Compressor Minimum OFF Time	After the compressor turns OFF, the controller shall prevent compressor restart until compressor_min_off_time has elapsed.
ACC-REQ-008	Compressor Activation	If cooling_demand == 1 and the compressor is not in minimum-OFF-time lockout, the controller shall set compressor_on = 1.
ACC-REQ-009	Compressor Deactivation	If cooling_demand == 0, power_button == 0, or a fault condition is active, the controller shall set compressor_on = 0.
ACC-REQ-010	Fan Activation During Compressor Activation	When compressor_on == 1, the controller shall set fan_on = 1.
ACC-REQ-011	Fan Behavior After Compressor Deactivation	When compressor_on == 0 after active cooling, the controller shall keep fan_on = 1 for configured fan_run_on_time, then set fan_on = 0.